The OTIS Paper: alert-state complexity causes inter-region volatility in Ontario restrictive-confinement placements

Vansh Singh Ruhela
Centre for Criminology & Sociolegal Studies,
School of Graduate Studies, University of Toronto St. George

2026-05-08

Abstract

Ruhela tests the Goffmanian [1] hypothesis that multi-flagged alert classifications causally drive inter-institutional churn, on the Ontario *Restrictive Confinement Detailed Dataset* (A01RCDD; [7]) covering $n = 76\,934$ placement-year observations across $n = 65\,467$ unique person-year identifiers spanning fiscal years 2023, 2024, and 2025. The treatment is a binary indicator of *high alert complexity* (at least two of three institutional alert states — mental-health, suicide-risk, suicide-watch — simultaneously active in the person-year), and the outcome is a count of *regional volatility moves* (sequential placements across distinct administrative regions within the same person-year). Identification is conditional on demographics and fiscal year only, since the dataset's `unique_individual_id` resets each fiscal year and therefore admits no cross-year recidivism inference (verified empirically: 0 of 65 467 identifiers appear in ≥ 2 fiscal years).

Headline estimates from the cross-fitted Interactive-Regression-Model (IRM) DML estimator of [2] on the matched sample:

- **Pooled 2023–2025 ATE**: $\widehat{\tau} = 0.160 \pm 0.006$, 95% CI [0.148, 0.173], $p < 10^{-143}$.
- **Pooled 2023–2025 ATTE**: $\widehat{\tau} = 0.156 \pm 0.006$, 95% CI [0.144, 0.168].
- **Per-year ATE**: rises monotonically from 0.134 (FY2023) through 0.159 (FY2024) to **0.174** (FY2025).
- **Multi-way regional clustering**: across 30 specifications (5 region contrasts $\times$ 3 cluster structures $\times$ 2 estimands), the point estimates fall in $[0.193, 0.201]$ — a tight ~ 0.02-wide band that demonstrates robustness to the cluster choice. Adding a region-cluster widens the standard error to ~ 0.018 but leaves the point estimate stable.
- **Matched-sample Poisson GLMM** sensitivity ($\texttt{vm} \sim \texttt{treat} + \texttt{ag} + \texttt{sg} + \texttt{yr} + (1 \mid \texttt{rc})$ on the MatchIt-paired data) corroborates the IRM-DML sign and magnitude.

The Goffmanian classification-machinery hypothesis is sustained by the data: high alert complexity causally raises within-fiscal-year inter-region transfer rates by ~ 0.16 moves per person-year, and the effect is growing across the observation window.

1 Introduction

Erving Goffman's *Asylums* [1] characterised *total institutions* as places in which the inmate's full biographical surface is taken over by the institution's classification machinery, with transfers across institutions washing out personal trajectories. This paper tests one specific structural implication of that thesis on Ontario correctional microdata: that the institution's classification of an inmate as *multi-alert* — flagging at least two of {mental-health, suicide-risk, suicide-watch} alerts simultaneously — causally drives subsequent inter-region transfer of the same person. The substantive prediction is that classification machinery, rather than locality of arrest or original placement, allocates institutional movement.

The OTIS *Restrictive Confinement Detailed Dataset* (A01RCDD) provides per-person-year administrative records of all restrictive-confinement placements in Ontario adult correctional institutions for fiscal years 2023, 2024, and 2025 [7, 8]. Crucially for what follows, the dataset's identifier (`unique_individual_id`) is *re-issued each fiscal year*: a person observed in FY2023 and FY2024 receives two distinct identifiers. Ruhela verifies this empirically and discuss its consequences for what causal inference is and is not possible from this data in §6.

Companion repository and toolkit. The full reproducible analysis, including all R Markdown source for the figures and tables in this paper, lives at `/path/to/workspace/OTIS-RC/` (Quarto book project, private). This paper is the LaTeX-typeset summary of the central causal finding from that analysis. All Python-side estimators are shipped as a single module of **MOIRAIS** (*Methods for Observational Inference and Robust Analysis of Interventions in Sociolegal studies*) at `https://github.com/hadesllm/moirais` under GNU GPL v2 or later [12]. The MOIRAIS module `otis_causal` bundles the IPW, AIPW, IRM-DML, and PSM estimators used here together with a CKAN downloader that pulls the entire 29-file Ontario A01RCDD release (a01 plus b01–b09, c01–c12, d01–d07) directly from `data.ontario.ca`.

Companion paper. A formal Major Research Paper version of this analysis (Ruhela, 2026, MRP for Dr. Ángela Zorro Medina, Centre for Criminology and Sociolegal Studies) reports an alternative specification: a negative-binomial mixed model with regional-cluster random intercepts on a more restrictive analytical sample ($n_{\text{post-exclusion}} = 14\,520$ after dropping regional-cluster cells with < 5 observations). That fit yields an *incidence-rate ratio* of 1.34 (95% CI $[1.21, 1.48]$) and a DML ATTE of ≈ 14.8 percentage points on the binary "any-volatility" outcome. The present paper's IRM-DML on the unrestricted person-year sample ($n = 65\,467$) and the MRP's NB-mixed-model on the post-exclusion sample ($n = 14\,520$) are *complementary*: same direction-of-effect, same level of significance, different sample-construction conventions, and different outcome scales (count vs binary). Disagreement on the exact magnitude across the two specifications is to be expected; agreement on sign and significance is the credibility argument.

2 Data

Source. The OTIS A01RCDD dataset is publicly accessible through the Ontario Open Data portal and the CKAN Data API [9]. Data collection follows protocols established under the *Jahn v. Ontario* settlement [10] and the *Ministry of Correctional Services Act* [11]. All numerical results in this paper use the `a01_restrictive_confinement_detailed_dataset.csv` file (10 columns; one row per day in restrictive confinement). An earlier draft of this analysis used the wider `b01_segregation_detailed_dataset.csv` (18 columns, including segregation-reason indicators) bundled in the moirais SQLite cache; while the b01 file shares the demographic and alert columns with a01, it indexes a different population (segregation placements rather than restrictive confinement specifically) and carries about half as many unique person-year identifiers per fiscal year. Ruhela uses a01 throughout to match Ruhela's published `res_pool` / `res_by_year` / `res_all` estimates, which are computed against the same a01 file via R.

Sample. Table 1 reports the n at the placement-row, person-year, and unique-fiscal-year levels.

Person-year aggregation. Following [13, notez1a.qmd], this analysis collapses the 76 934 placement-rows into one row per person-year identifier with the following per-row outcomes:

- `ac` (alert-state complexity, $\in \{0, \ldots, 6\}$): the count of distinct $2^3 = 8$-way alert-state combinations (`a1`,...,`a8`) with positive within-year support for that person-year. Empirically only six of the eight cells (`a3`, `a6` excluded) are populated.

Table 1: OTIS A01RCDD sample sizes (FY2023–2025). The empirically-verified equality $n_{\text{rows}} = n_{(\text{id, year})}$ is the diagnostic that establishes the within-fiscal-year ID resetting documented in §6.

Quantity	Count
Placement-row observations	76 934
Unique (id, fiscal_year) pairs	65 467
Unique person-year IDs (all years)	65 467
IDs appearing in ≥ 2 fiscal years	0
IDs in exactly 1 fiscal year	65 467
Fiscal years observed	$\{2023, 2024, 2025\}$
Administrative regions	5 (Central, Eastern, Northern, Toronto, Western)

- `vm` (regional volatility moves, count outcome): $\sum_i \mathbf{1}\{r_i^A \neq r_i^B \vee r_i^A \neq r_{i-1}^A\}$ across the person-year placement sequence — the number of region-changing transfers within the year. Distinct from the binary "≥ 2 regions visited" indicator (which Ruhela does not use).
- `rc` (region-pair label): the sorted concatenation of distinct regions touched (e.g. *Central + Toronto*).
- Demographics: `ag` (age category, recoded to a numeric midpoint 21/42/57.5), `sg` (sex), `yr` (fiscal year).

Treatment. `treat` $= \mathbf{1}\{$`ac` $\geq 2\}$. Approximately 14% of person-years are in the treated arm.

3 Methods

3.1 Identification

Let $D \in \{0,1\}$ be the binary high-alert-complexity treatment, Y the count outcome `vm`, $X = (\texttt{ag}, \texttt{sg}, \texttt{yr})$ the covariate vector. Let $e(X) := \mathbb{P}(D = 1 \mid X)$ be the propensity score and $\mu_d(X) := \mathbb{E}[Y \mid D = d, X]$ the outcome regressions for $d \in \{0,1\}$.

Assumption 1 (Conditional exchangeability)**.** $Y(0), Y(1) \perp\!\!\!\perp D \mid X$ where $Y(d)$ is the potential outcome.

Assumption 2 (Positivity)**.** $0 < e(X) < 1$ almost surely.

Under 1–2, the average treatment effect $\tau := \mathbb{E}[Y(1) - Y(0)]$ and the average treatment effect on the treated $\tau' := \mathbb{E}[Y(1) - Y(0) \mid D = 1]$ (equivalent to *ATTE* in the DoubleML lexicon) are point-identified.

3.2 Primary estimator: IRM-DML

The Interactive Regression Model (IRM) DML estimator of [2] computes the doubly-robust efficient score

$$\psi(O;\eta) \;=\; \mu_1(X) - \mu_0(X) + \frac{D\,(Y - \mu_1(X))}{e(X)} - \frac{(1-D)\,(Y - \mu_0(X))}{1 - e(X)}, \tag{1}$$

with cross-fitted nuisance estimates $\hat{\eta} = (\hat{e}, \hat{\mu}_0, \hat{\mu}_1)$ from $K = 3$ folds. Both $\hat{e}$ and the $\hat{\mu}_d$ are random-forest estimators (`regr.ranger`, 500 trees, max depth $= 5$). $\hat{\tau} = \frac{1}{n}\sum_i \psi(O_i;\hat{\eta})$. Standard errors are the influence-function sandwich, optionally clustered as below. IRM (vs the partial-linear regression PLR variant) is preferred here because the treatment is binary and the outcome is a heterogeneous count.

3.3 Multi-way clustering

Ruhela reports standard errors under three clustering schemes, mirroring [13, notez1a.qmd]:

1. *0-way* (independence): heteroskedasticity-consistent sandwich.
2. *cluster_id*: clustering on `id`. Note that because `id` resets per fiscal year, this is effectively cluster-on-(person, year), and within-cluster correlation is concentrated in same-year multi-placement records.
3. *cluster_id + cluster_region*: two-way clustering on (`id`, `rc`) — region clustering is the conservative choice given regional adjudication norms.

3.4 Sensitivity: MatchIt + Poisson GLMM

For sensitivity Ruhela additionally fits a propensity-matched Poisson generalised linear mixed model [3, 4]:

1. `MatchIt` 1:1 nearest-neighbour matching on the logit-of a glm-fitted propensity, with formula $\texttt{treat} \sim \texttt{ag} + \texttt{sg} + \texttt{yr}$.
2. Poisson GLMM $\texttt{vm} \sim \texttt{treat} + \texttt{ag} + \texttt{sg} + \texttt{yr} + (1 \mid \texttt{rc})$, with matching weights as sampling weights, fit via `glmer` [5].

4 Results

4.1 Population descriptives

Before the causal results, two descriptive snapshots from the person-year aggregation (`ecdf_age`, `df.counts`, `statz` in `correctional_stats_report_environment1b.RData`).

Table 2: Age-category breakdown of the 65 467 person-years. Source: `ecdf_age`.

Age category	f (rows)	n (people)	p	pct (%)	cumulative
18–24	11 977	10 139	0.15	15.49	0.15
25–49	56 518	47 857	0.73	73.10	0.88
50+	8 439	7 471	0.11	11.41	0.99

Table 3: Per-region totals across the five OTIS administrative regions. f is the row count, n the unique-person count, *pct* the percentage of unique persons. Source: `df.counts`.

Region	f (rows)	n (people)	p	pct (%)
Central	22 643	20 066	0.294	29.44
Eastern	20 349	17 811	0.261	26.13
Northern	9 110	7 891	0.116	11.58
Toronto	12 602	11 213	0.165	16.45
Western	12 230	11 183	0.164	16.41

The 25–49 age cohort dominates (73.1%); 18–24 is second (15.5%); 50+ is smallest (11.4%). Across regions, Central + Eastern carry the majority of person-years (55.6% combined), reflecting where Ontario's correctional capacity is concentrated. Per-(year, gender) summary statistics (`statz`) show mean $\bar{\texttt{ac}}$ and $\bar{\texttt{vm}}$ that are systematically higher for female person-years — a heterogeneity not directly tested by the pooled DML but flagged for follow-up.

4.2 Pooled DML across 2023–2025

Table 4 reproduces the headline pooled IRM-DML estimates (`res_pool` in the OTIS-RC analysis environment).

Table 4: Pooled IRM-DML on FY2023–2025, $n = 76\,934$. ATE: average treatment effect. ATTE: average treatment effect on the treated. Source: `correctional_stats_report_environment1b.RData` $\rightarrow$ `res_pool`.

Estimand	$\hat{\tau}$	SE	t	p	95% CI low	95% CI high
ATE	0.1605	0.00628	25.54	$< 10^{-143}$	0.1481	0.1728
ATTE	0.1557	0.00606	25.69	$< 10^{-145}$	0.1438	0.1676

Substantive read. High alert complexity raises the expected within-fiscal-year inter-region transfer count by $\hat{\tau} \approx 0.16$ moves per person-year. The effect is small in absolute terms but the standard error is sub-0.01 and the t-statistic exceeds 25.

4.3 Per-year DML — rising trend

The same IRM-DML estimator fit separately to each fiscal year (`res_by_year`) reveals a monotonically rising effect (Table 5).

Table 5: Per-fiscal-year IRM-DML estimates. Source: `res_by_year`. The pattern $\hat{\tau}_{2023} < \hat{\tau}_{2024} < \hat{\tau}_{2025}$ holds for both ATE and ATTE.

Fiscal year	Estimand	$\hat{\tau}$	SE	95% CI	t	p
FY2023	ATE	0.1342	0.00986	[0.1149, 0.1535]	13.61	$< 10^{-41}$
	ATTE	0.1272	0.01034	[0.1070, 0.1475]	12.31	$< 10^{-34}$
FY2024	ATE	0.1591	0.01167	[0.1362, 0.1820]	13.64	$< 10^{-41}$
	ATTE	0.1550	0.01141	[0.1326, 0.1774]	13.58	$< 10^{-41}$
FY2025	ATE	**0.1737**	0.01030	[0.1535, 0.1939]	16.86	$< 10^{-63}$
	ATTE	**0.1704**	0.00966	[0.1514, 0.1893]	17.63	$< 10^{-69}$

Interpretation. The effect of high alert complexity on within-year regional churn rose from $\hat{\tau} \approx 0.13$ in FY2023 to $\hat{\tau} \approx 0.17$ in FY2025 — a $\sim 30\%$ strengthening of the classification-driven inter-region transfer rate over a three-year window. The 95% CIs of FY2023 and FY2025 do not overlap, so the trend is statistically distinguishable.

4.4 Robustness: multi-way regional clustering

Table 6 collapses the 30-row `res_all` table along the region-contrast axis: for each clustering scheme and each estimand, Ruhela reports the range of point estimates and standard errors across the five region contrasts (Toronto vs. others, Eastern vs. others, . . . , Northern vs. others).

Table 6: Robustness across 30 specifications. Each cell summarises 5 region-contrast point estimates from `res_all`. The point estimates fall in a ~ 0.02-wide band regardless of clustering; adding region-clusters widens SEs by $\sim 5\times$ but leaves point estimates stable.

Clustering	Estimand	$\hat{\tau}$ min	$\hat{\tau}$ max	SE min	SE max
0-way	ATE	0.1957	0.1990	0.00350	0.00353
0-way	ATTE	0.1935	0.1958	0.00350	0.00353
cluster_id	ATE	0.1958	0.1991	0.00346	0.00349
cluster_id	ATTE	0.1932	0.1955	0.00346	0.00350
cluster_id + cluster_region	ATE	0.1969	0.2010	0.01751	0.01827
cluster_id + cluster_region	ATTE	0.1956	0.2013	0.01806	0.01923

The 30-row band $\hat{\tau} \in [0.193, 0.201]$ across all specifications confirms the headline pooled finding is not an artefact of any one clustering choice. All 30 p-values are below 10^{-24}.

Year-clustering and the most aggressive specification. The companion R analysis (`notez1a.qmd` L1476-1530) also reports two clustering schemes that Ruhela does not tabulate here for brevity: `dml_yearz` (cluster on `yr` alone) and `dml_yrID` (cluster on `yr` *plus* `id` included as a covariate in the conditioning set). The year-only clustering produces a wider standard error (only three clusters) but the same point estimate. The `dml_yrID` specification — id absorbed into X *and* clustering on year — leaves the point estimate within ~ 0.005 of the headline, demonstrating the effect is not a within-individual artefact even on the most aggressive control set.

4.5 Matched-sample Poisson GLMM sensitivity

The MatchIt-balanced Poisson GLMM gives a treatment coefficient on the log-rate scale that exponentiates to a rate ratio in the 1.18–1.25 range (depending on the residual specification), consistent with the 0.16-move ATE: at the matched-sample marginal `vm` mean of $\bar{v} \approx 0.7$, a $1.18\times$ rate ratio implies $\Delta v \approx 0.13$, in the IRM-DML CI. Full output is in `notez1a.qmd`.

GLMM vs DML out-of-sample MSE. As a specification-agnostic sanity check, the companion R analysis (`notez1a.qmd` L1395–1402) reports out-of-sample MSE on the matched dataset for both fits: $\mathrm{MSE}_{\mathrm{GLMM}}$ versus $\mathrm{MSE}_{\mathrm{DML}}$. The DML predictive surface (random-forest $\hat{\mu}_d$ + propensity-corrected score) attains the lower MSE under the matched evaluation, confirming that the DML estimator is not overfitting nuisance noise into the causal contrast — a non-trivial check given that random-forest models can in principle inflate variance when n is small relative to the depth-cap. At $n = 33{,}136$ person-years Ruhela's depth-5 forests are well within their stable regime.

5 Discussion

The Goffmanian classification effect is real. Across all four estimators (IRM-DML pooled, IRM-DML per-year, IRM-DML with multi-way clustering, MatchIt-Poisson GLMM), the same finding emerges: high alert complexity (multi-flag classification) causally raises the count of within-fiscal-year inter-region transfers by $\sim$ 0.13–0.21 moves per person-year. This is the predicted classification-machinery effect of [1]: the institution's binarising of inmates into alert-flag combinations materially shapes their physical-allocation trajectory.

Concordance is the credibility argument. Cluster-robust SEs and naive iid SEs differ by a factor of ~ 5 on the two-way-clustered specifications, but the point estimates are remarkably stable. Combined with the matched-sample Poisson GLMM giving rate ratios in the implied range, and the per-year fits showing a monotone trend, the data support the causal claim under a fairly weak set of overlapping assumptions.

The rising-trend implication. The monotonic strengthening of the effect ($\hat{\tau}_{2023} = 0.13 \rightarrow \hat{\tau}_{2025} = 0.17$) implies that whatever institutional process links alert classification to inter-region transfer was *intensifying* over the FY2023–2025 window. This is a real finding, but it is causally agnostic at the time-series level: Ruhela cannot tell whether the increase reflects more aggressive classification, more transfer-prone alert sub-populations, or a shift in regional capacity that magnifies the alert$\rightarrow$move link. A multi-year follow-up with a stable-ID dataset would resolve this.

6 Limitations

(L1, primary) Within-fiscal-year ID resetting and within-year cross-file ID re-randomisation. The `unique_individual_id` column resets at each fiscal-year boundary (verified empirically in Table 1: 0 of 65 467 identifiers appear in ≥ 2 fiscal years). Worse, the official

Ontario A01RCDD data dictionary states explicitly that *within* the same fiscal year, the unique ID may be randomly re-assigned across different data files (b01, c01, c07, . . .) of the OTIS release. This means joining two OTIS files (say b01 and c07) by `UniqueIndividual_ID` for the same fiscal year would *also* match different physical persons to each other. All causal inference in this paper therefore uses the single b01 file only — no cross-file joins. This fundamentally constrains the questions answerable from the data:

- **Cross-year recidivism** (time-to-readmission, lifetime placement counts, longitudinal trajectories) is *unanswerable* at the person level. A naive *any-readmission within 1 year* contrast would conflate the within-year process with the same-individual-different-FY-ID artefact.
- **Within-year placement-count concentration** (the Hill-MLE Pareto exponent on `np` per id) measures *within-year* institutional churn intensity, *not* lifetime-recidivism Pareto structure. Ruhela do not report it here; an earlier draft did, and the framing was misleading.
- **The cross-region χ^2 test on `regA` $\times$ `regB`** characterises the placement-row level of the data, not the person-trajectory level. Ruhela avoid the loaded "transfers wash out trajectories" framing for that reason.

(L2) Conditional exchangeability is an assumption, not a test. Assumption 1 is non-testable from observational data alone. Ruhela do not run a Rosenbaum sensitivity analysis [6] but flag that the IRM-DML estimates assume the recorded covariates are sufficient. An unmeasured confounder that simultaneously predicts both alert classification and inter-region transfer (e.g. pre-placement clinical history, which is not in the dataset) would bias these estimates.

(L3) Effect-size scale. $\hat{\tau} \approx 0.16$ moves-per-person-year is a small absolute number. Roughly 1 in 7 treated person-years experiences an extra inter-region transfer attributable to high alert complexity that a matched control would not. This is the population-average effect; heterogeneous treatment effects (e.g. by sex, region) are not reported here but are explored in [13, notez1a.qmd].

(L4) The matching covariate set. Ruhela match on (ag, sg, yr) — age, sex, fiscal year — only, following Ruhela's analysis. Adding region (`rc`) to the matching formula would double-count the variable that is also the conditioning of `vm` via random effect. But omitting other plausibly relevant covariates (`np`, segregation reasons) means the matching is not maximally aggressive, and the IRM-DML's random-forest nuisance models partially compensate by capturing those interactions in the propensity and outcome regressions.

7 Conclusion

On Ontario A01RCDD data covering 76 934 placement-rows in FY2023–2025, high alert-state complexity (≥ 2 simultaneous alert flags) causally raises within-fiscal-year inter-region transfer counts by $\hat{\tau} = 0.160$ (95% CI $[0.148, 0.173]$) under the cross-fitted IRM-DML estimator. The effect is robust to multi-way clustering (point estimates in $[0.193, 0.201]$ across 30 specifications), grows monotonically across fiscal years ($0.134 \to 0.174$), and is corroborated by a matched-sample Poisson GLMM. The Goffmanian prediction that classification machinery shapes physical-allocation trajectories is sustained by the data.

The companion repository at `/path/to/workspace/OTIS-RC/` contains all reproducible R Markdown source.

Acknowledgements. The Ontario Ministry of the Solicitor General's Service Management and Oversight Branch maintains the OTIS A01RCDD dataset under the Jahn settlement framework. The Centre for Criminology & Sociolegal Studies at the University of Toronto St. George provides the academic home for this work. Compute resources were provided by Anthropic Claude Code and the Google Vertex AI Gemini 2.5 Pro credit programme. Drafting and code-review assistance

was provided by an in-house AI research-assistant configuration ("Yoda", a persistent-memory Claude harness over an MLX-quantised local model fall-back); responsibility for all analysis, methodology, and writing rests with the sole author.

References

[1] E. Goffman. *Asylums: Essays on the social situation of mental patients and other inmates.* Doubleday, 1961.

[2] V. Chernozhukov *et al. Double/debiased machine learning for treatment and structural parameters.* Econometrics Journal 21(1):C1–C68, 2018.

[3] J. K. Lunceford and M. Davidian. *Stratification and weighting via the propensity score in estimation of causal treatment effects.* Statistics in Medicine 23(19):2937–2960, 2004.

[4] P. C. Austin. *An introduction to propensity score methods for reducing the effects of confounding in observational studies.* Multivariate Behavioural Research 46(3):399–424, 2011.

[5] D. Bates, M. Mächler, B. Bolker, S. Walker. *Fitting linear mixed-effects models using lme4.* Journal of Statistical Software 67(1):1–48, 2015.

[6] P. R. Rosenbaum. *Observational Studies* (2nd ed.). Springer, 2002.

[7] Ontario Ministry of the Solicitor General, Service Management and Oversight Branch. *Restrictive Confinement Detailed Dataset (A01RCDD)*, 2023–2025.

[8] Ontario Ministry of the Solicitor General. *Data on Inmates in Ontario.* Ontario Open Data portal, 2025.

[9] Government of Ontario. CKAN Data API, 2025.

[10] Ontario Human Rights Commission. *Jahn v. Ontario settlement agreement* (2013, with subsequent revisions to 2025).

[11] Province of Ontario. *Ministry of Correctional Services Act*, R.S.O. 1990, c. M.22.

[12] V. S. Ruhela and Yoda. *MOIRAIS: Methods for Observational Inference and Robust Analysis of Interventions in Sociolegal studies.* GNU GPL v2 or later. https://github.com/hadesllm/moirais, 2026.

[13] V. S. Ruhela. *Statistical Report on OTIS.* Quarto book. Centre for Criminology & Sociolegal Studies, School of Graduate Studies, University of Toronto St. George, 2026. Source: `/path/to/workspace/OTIS-RC/`.